

Development of statistical methods for vibration analysis and consideration of mechanical environments - Application to land vehicles

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Abstract:

Wheeled land transport vehicles and their on-board equipment are subject to vibrations during use, particularly during travel. Due in particular to the irregularity of rolling surfaces, these stresses are of a non-Gaussian nature. In order to assess the effects of vibratory phenomena on systems, and to enable the construction of qualification tests, a method for taking account of mechanical environments is described in the AFNOR X50-144-3 edition 2 standard. It consists in particular in processing samples from measured data by stationarity class, then inferring the statistical distributions obtained. This contribution presents a number of developments in this approach. A new data classification method is proposed, based on a spectral content similarity criterion. New “a priori” statistical laws are implemented for the inference of statistical distributions from measurements over short timescales, in order to propagate them to the scale of the system life profile, over long durations. The relevance of the methods introduced is assessed by applying them to real data from measurements, as well as to digitally generated signals with known spectral characteristics.

Keywords: Vibration, Environment, Mechanic, Tailoring process, AFNOR X50-144

1 Introduction

During their lifetime, wheeled ground transportation vehicles and their onboard equipment are subjected to stresses generated by the environment. The vibratory phenomena encountered are non-stationary and non-Gaussian, mainly due to the irregularity of the rolling surfaces and the shocks encountered. To account for the effects of the mechanical environment and evaluate the performance of systems, various methods are documented. Having supplanted spectral approaches, deterministic processing methods, however, prove to be non-conservative for predicting non-stationary and non-Gaussian phenomena [1]. The AFNOR standard X50-144-3 edition 2 [2] replaces the GAM-EG-13 [3], which was previously used for the qualification of military ground equipment. It addresses the need to account for the stochastic nature of loading conditions [4, 5]. It outlines a tailoring approach to account for mechanical environments, evaluate the effects of vibratory phenomena on systems, and allow for the design of qualification tests. This includes processing by stationarity class of the samples from the measured data and inferring the statistical distributions. However, the spectra thus obtained

show significant fluctuations and are not evaluated in comparison with historical processing methods, particularly for Gaussian signals [6]. The present contribution therefore aims to address these limitations.

2 Maximum Response and Fatigue Damage Spectra

Before addressing the statistical processing methods of vibratory data, it is important to recall some concepts related to the consideration of environmental effects through response spectra. The system under study is modelled by a set of one-degree-of-freedom (dof) resonators with mass m , stiffness k , constant damping c , and a natural frequency f_0 within the range $[f_0^{\min}; f_0^{\max}]$, covering the zone of its sensitive frequencies (figure 1). The failure modes are the exceedance of the admissible threshold stress and rupture due to accumulated fatigue under the effect of alternating stress cycles.

The Maximum Response Spectrum (MRS) and the Fatigue Damage Spectrum (FDS) are indicators of damage induced by the considered failure modes. They are calculated from a time-domain excitation $x(t)$ applied to the input of the considered one-dof resonator.

The deterministic MRS characterizes the local extrema $Z_{\max}$ of the responses $z(t)$ through the resonator. It is defined as follows [2].

$$MRS_d(f_0) = 4\pi^2 f_0^2 Z_{\max}(f_0) \text{ with } Z_{\max} = \text{Max}[|z(t)|] \text{ for } t \in [0; T]$$

At the stochastic point of view, the MRS is defined as follows, where $Z_{\max, \alpha}$ is the quantile of the largest overall extrema of the relative displacement calculated from its distribution function $F_{Z_{\sup}}(z)$ for an exceedance risk α over the stress duration T .

$$MRS_{\alpha} = 4\pi^2 f_0^2 Z_{\max, \alpha}(f_0) \text{ with } Z_{\max, \alpha} = F_{Z_{\sup}}^{-1}(1 - \alpha)$$

The FDS represents the accumulation of fatigue damage in the system. It is calculated based on the Basquin's law ($N\sigma^b = C$, see figure 1) and the linear law of Miner [7]. From a deterministic point of view, it is defined, for the resonator with natural frequency f_0 and the duration of loading T , by the following formula [2].

$$FDS_d(f_0) = \sum_i \frac{n_i}{N_i} = \frac{K^b}{C} \sum_i n_i (z_i, \xi, f_0)^b$$

where C is the constant in Basquin's law, b is the slope of Basquin's law modelling the Wöhler curve associated with the one-degree-of-freedom system, z_{i, ξ, f_0} is the i -th loading cycle of the relative displacement $z(t)$ with duration T , ξ is the damping ratio of the resonator, and N_i is the number of cycles at 50 % failure under the stress level σ_i . The counting of alternating cycles can be performed using the Rainflow method [8]. In the stochastic sense, the FDS is defined as follows.

$$FDS_{\alpha} = Dc_{\alpha}(f_0) \text{ with } Dc_{\alpha} = F_{Dc}^{-1}(1 - \alpha)$$

where Dc_{α} is the quantile of the cumulative damage associated with the relative displacement, calculated from its distribution function $F_{Dc}(d)$ for an exceedance risk α over the duration of the load T .

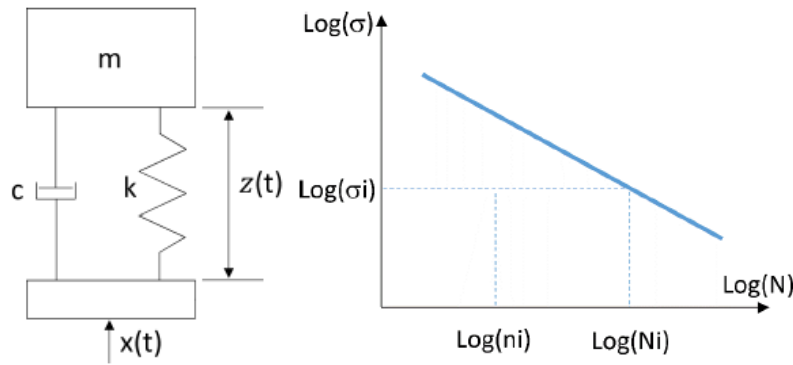


Figure 1 One-DOF response model (left) and endurance curve – Basquin model (right). σ is the stress level and N is the number of cycles to failure.

Other approaches are documented in the literature, such as analytical methods for processing Gaussian signals [9, 10, 11] and time-domain analysis methods for non-Gaussian signals [12].

3 Disjoint Blocks Method processing according to the standard X50 144-3 ed. 2

The X50 144-3 ed. 2 standard from January 2021 [2] describes in its annex C the methodology for processing any stochastic vibratory signal in order to extract the response spectra. Compared to the tools available until then, the standard innovates by determining statistical distributions from the measured data. A method for extrapolation to long times, based on the stochastic synthesis of distributions, is proposed. This allows for a renewal of life profile construction techniques.

For clarity, the hypothesis is made that the signal to be processed is stationary. The consideration of non-stationarity is addressed in paragraph 4.1. The processing technique is based on the Disjoint Block Method (DBM). This method consists of dividing the signal into n segments of constant duration t_B , then calculating for each resonator with frequency f_0 the MRS and the FDS for each of the n time samples (see [2] annex C § C3). The choice of the blocks duration t_B , in the so-called “indirect” DBM dedicated to real signals analysis, is a trade-off between the need to have a sufficiently large number of samples, which leads to reducing t_B , and the need to have a sufficient number of maxima related to the resonator period, which leads to increasing t_B . Therefore, the lowest frequency of the analysis range conditions the duration t_B . In most cases, this value is of the order of a few seconds.

The strategy for processing the n -samples differs depending on whether we are dealing with the MRS or the FDS. In the case of n -samples of MRS, the standard [2] proposes performing statistical inference based on the L-moment method [13], a technique well-suited to small sample sizes. For n -samples of FDS, the standard describes an approach identical to that used for n -samples of MRS. In practice, this step is unnecessary. Indeed, a simple calculation of the mean and variance performed on the n -sample is sufficient; the Central Limit Theorem (CLT) guarantees the convergence of the cumulative fatigue process to a Gaussian distribution, regardless of the nature of the distribution of the n -sample of FDS. It is specified that the mean and variance of the process extrapolated over a life profile duration are proportional to the ratio $N = t_p/t_B$, where t_p represents the extrapolated duration of the life profile.

Only the issue of MRS is addressed in this contribution.

3.1 Statistical inference of MRS according to the standard X50 144-3 ed.2

The statistical inference of a n-sample of MRS relies on the knowledge of the analytical expression of the L-moments, derived from the linear combination of the weighted probability moments. These expressions are only known for some specific statistical distributions [13]. Similarly, the L-moments can be evaluated experimentally from the numerical values of the n-sample. It is then possible to calculate the parameters of the statistical distribution.

The standard [2] proposes 7 distributions referred to as “a priori distributions or APD”:

- 3 extreme value distributions (Gumbel with 2 parameters, Fréchet, and negative Weibull with 3 parameters)
- 2 Weibull distributions (2 and 3 parameters)
- 2 log-normal distributions (2 and 3 parameters)

The properties of the different distributions are provided in table C3 of annex C of the standard [2].

After identifying the parameters of each distribution, a comparison of the different APD cumulative distribution functions with the experimental cumulative distribution function is carried out. To choose the most suitable distribution, the Mean Square Deviation Index (MSDI) is defined below.

$$MSDI(\%) = \frac{100}{m} \sum_{i=n-m+1}^n \left(\frac{X_{i:n}^{exp} - X_{i:n}^{LAP}}{X_{i:n}^{exp}} \right)^2$$

The APD with the lowest MSDI is retained, becoming the Distribution To Use (DTU). The comparison is only carried out on the m experimental points beyond the mean value on the right tail of the distribution, in order to prioritize the adjustment of the distribution's tail.

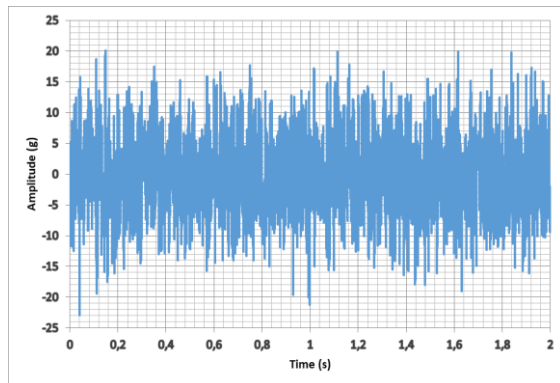
The choice of the 7 APD was primarily guided by the knowledge of the analytical expressions of the L-moments, rather than by industry-specific considerations. Thus, the generalized Rayleigh distribution, which is by definition the DTU for Gaussian vibratory signals [9, 14], was not included because the analytical expressions of its L-moments were not available at the time the standard [2] was published.

Once the DTU is defined, the extrapolation of the MRS variable behaviour from the observation time t_B to the life profile time t_p is carried out. According to the theory of random variable propagation, the MRS variable over a time t_p follows a cumulative distribution function of the form $(DTU)^N$, where $N = t_p/t_B$. According to extreme value theory, the function $(DTU)^N$ asymptotically converges, over long observation times (consistent with those of a life profile), to one of the three Extreme Value Distributions (EVD) of the Gumbel, Fréchet, or Negative Weibull type. For each of the 7 proposed DTU, it is possible to calculate the parameters of the asymptotic distributions (see [2] annex C table C1), and thus determine the value of the MRS quantile corresponding to the exceedance risk α .

3.2 Limitations of the method

When the standard [2] was published in 2021, an experimental validation study of the method was initiated by a group of experts within the framework of the mechanical and climatic commission of the Association for the Development of Environmental Sciences and Techniques (ASTE). This study aimed to demonstrate the validity of the DBM applied to a signal whose stochastic MRS was known analytically. To do this, a Gaussian time signal was generated from a Power Spectral Density (PSD) using a control console over a period of 600 seconds. The properties of the signal are given in figure 2. The definition of the cumulative distribution function of the MRS of a zero-mean Gaussian excitation

signal is known and documented in the case of a lightly damped resonator ($<5\%$) (see [2] annex A). It is thus shown that the MRS variable follows a generalized Rayleigh distribution asymptotically tending toward a Gumbel distribution for long times. The excitation duration was set to 10 000 hours in the case studied. The inference method using DBM is implemented over a duration t_B of 1 second, with the size of the n-samples for each degree of freedom being 600.

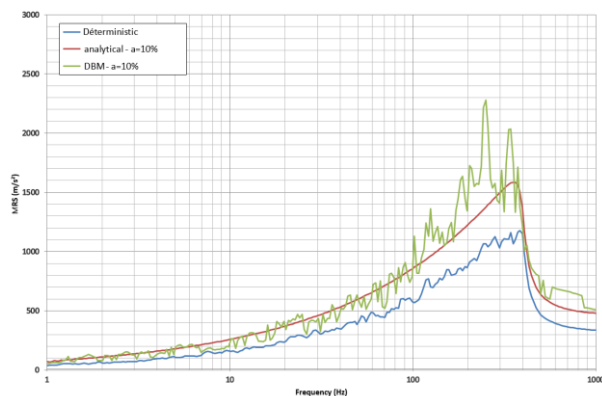


Nature of the signal: Gaussian random noise without clipping
 Level: White noise 0.1 g²/Hz from 1 to 400 Hz
 Frequency resolution: 0.125 Hz (3200 spectral lines)
 Block duration: 8 seconds (1/0.125)
 Signal sampling frequency: 12 800 Hz
 Signal duration: 600 seconds
 Control system: Data Physics
 Generation: Closed-loop control

Figure 2 Amplitude as function of time (extract) and properties of the generated excitation signal.

The MRS obtained from the analytical calculation, serving as reference for $\alpha=10\%$, the projection obtained by the DBM at the same risk, and the curve derived from the deterministic calculation over the entire generated time signal are plotted in figure 3.

The table in figure 3 summarizes the relative deviations calculated compared to the analytical reference.



	DBM vs Analytical	Deterministic vs Analytical
Mean deviation	2 %	-34 %
Max deviation	66 %	-16 %
Min deviation	-33 %	-49 %

Figure 3 Deterministic MRS (blue), analytical MRS (red), and MRS obtained using DBM (green) at the risk of exceedance $\alpha=10\%$ (left). Analysis of the relative deviations of the MRS calculated using different methods (right).

The deterministic MRS is by definition consistently below the analytical reference, with an average relative deviation of -34 %. The MRS obtained by DBM for a duration of 10 000 hours shows an average relative deviation of 2 % compared to the reference, but a maximum deviation of 66 % near 262 Hz. Thus, the method seems relevant for projecting the MRS but presents erratic fluctuations between the degrees of freedom, which affect the precision.

The cause of these abrupt variations that impact the MRS calculation is explained by the process of constructing the cumulative distribution function associated with the excitation time of the life profile. Indeed, the observation time t_B of the n-sample allows the determination of the DTU by inference

using the method described earlier, and the extrapolated cumulative distribution function for the life profile duration t_p is $(DTU)^N$. This phenomenon is graphically illustrated in figure 4. The DTU is determined for values of the MRS variable within the "calibration zone". The extrapolation technique causes, for $N > 1$, a "shift" of the cumulative distribution function to the right. The "usage zone" of the life profile cumulative distribution function is thus built with the right tail of the DTU, which has not been recalibrated. As a result, at different frequencies, DTU may perfectly interpolate the experimental cumulative distribution function in the calibration zone but behave very differently in the usage zone, that explains the significant variations observed on the curve presented in figure 3.

The conclusion of this experimentation is that proposing a multi-choice APD seems to be not an appropriated way. A suitable answer would be probably to find a unique APD able to match as good as possible the signal process.

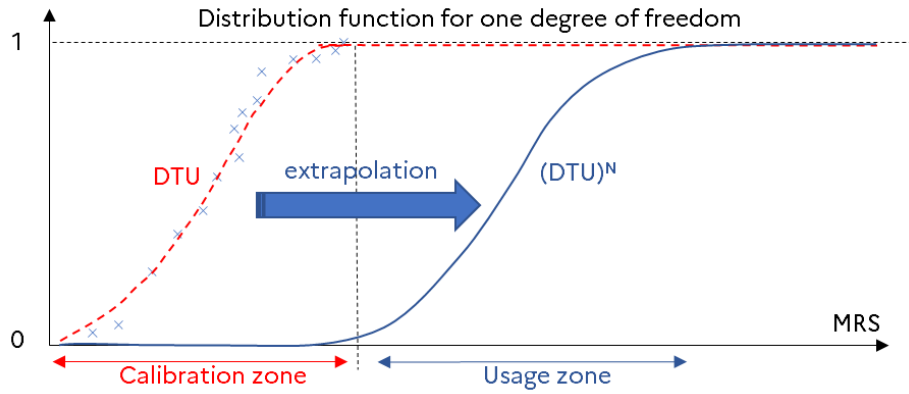


Figure 4 Extrapolation of the cumulative distribution function of the MRS for the life profile.

4 Methods

4.1 Data classification by spectral similarity

Standard [2] has not yet recommended a robust classification method that takes into account the spectral content of the signals. A new method for classifying stochastic vibratory signals in terms of spectral stationarity is introduced. To achieve this, the considered excitation signal is divided into temporal blocks of finite duration. These blocks are of size 2^n to be compatible with Fast Fourier Transform (FFT) processing. To compare the stationarity of the formed blocks, a spectral similarity index (SSI) $ssi_{i,j}$ is defined as follows.

$$ssi_{i,j} = 1 - \frac{\int_{f_{min}}^{f_{max}} |\log(\text{spectrum}_j(f)) - \log(\text{spectrum}_i(f))| \cdot df}{\int_{f_{min}}^{f_{max}} \log(\text{spectrum}_i(f)) \cdot df}$$

The parameters of the SSI formulation undergo a logarithmic transformation before being integrated, in order to ensure spectral similarity across the entire frequency range studied. The calculation of the SSI between each of the spectra forms the SSI matrix. The clusters are then formed by grouping the blocks whose stationarity difference is above the spectral similarity threshold. The lower the threshold value, the greater the stationarity within the cluster, but the higher the number of clusters and the smaller their size. The goal is to find a balance that forces spectral similarity while limiting the number

of samples. In practice, feedback shows that a threshold value of 85 % is relevant for the targeted applications. The principle of stationary clustering of the excitation signal is summarized in figure 5. If the studied signal is stationary, a single cluster is formed after classification based on spectral similarity.

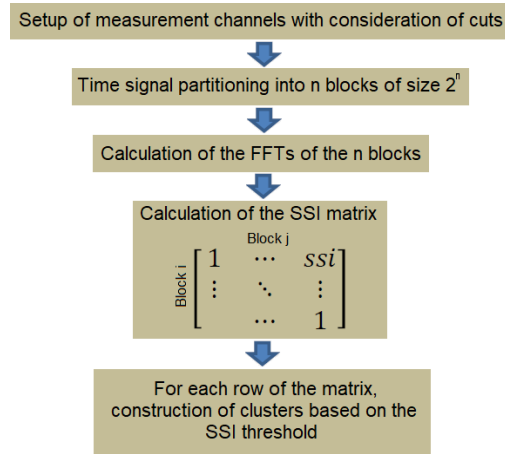


Figure 5 Principle of partitioning the excitation signal into stationary clusters.

4.2 Statistical models

In order to improve the DTU determination by using a unique APD, and thus correct the phenomena observed in the experimentation of §3.2, new models are implemented, including the Kappa distribution with four parameters and generalized Rayleigh distribution.

Initially introduced by Hosking and Wallis [13], the Kappa distribution is particularly applied to the prediction of extreme events in the field of hydrology [15]. It is defined, for $\xi \in R$, $\alpha > 0$, $k > -1$ and $(h.k > -1 \text{ if } h < 0)$ by the following cumulative distribution function [13].

$$F(x) = \left[1 - h \cdot \left\{ 1 - k \cdot \frac{(x - \xi)^{1/k}}{\alpha} \right\} \right]^{1/h}$$

The Kappa distribution has also the particularity to get an asymptotic convergence to the 3 EVD. To study the distribution law of the variables of interest, the present work primarily relies on the ratios of the L-moments τ_3 (L-skewness) and τ_4 (L-kurtosis) [13]. Thus, the validity domain of the Kappa distribution law is defined by its lower (red) and upper (blue) limits on the $(\tau_3; \tau_4)$ diagram in figure 6. The a priori Kappa distribution has the particularity of having its statistical model of the random variable asymptotically tend towards EVD. The domains of definition of these laws also appear in figure 6. The domain of the Gumbel distribution is the junction point of the curves representing the Weibull (green) and Fréchet (purple) distributions. The curve $k=0$ (brown) establishes the separation between the domains covered by Weibull like ($k>0$) and Fréchet like ($k<0$) distributions.

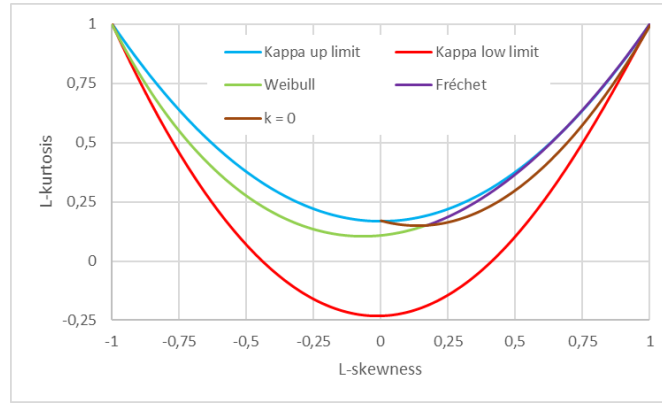


Figure 6 Diagram of the L-moment ratio ($\tau_3 ; \tau_4$) for different statistical models.

The generalized Rayleigh distribution is also implemented in the present work to model the stochastic nature of the maximum stress response. Indeed, this distribution is very popular in the environmental domain being the exact model of Gaussian excitation. It is defined, for $\alpha > 0$ and $\lambda > 0$, by the following cumulative distribution function [16].

$$F(x; \alpha, \lambda) = (1 - e^{-(\lambda x)^2})^\alpha ; x > 0$$

The expressions for the L-moments of the generalized Rayleigh distribution are given by Kundu and Raqab [16] through the change of variable $X = x^2$, based on the L-moments of the generalized exponential distribution.

4.3 Non Gaussian signal generation

In order to validate the proposed methods, they are applied to excitation signals whose effects are known. To do this, non-Gaussian signals are numerically generated from a PSD and a kurtosis [17, 18], and their response spectra are obtained from the converged data [19].

Thus, starting from the target PSD, a random phase generation and an inverse FFT are performed to obtain a Gaussian excitation signal. The generated signal is then scaled to unit variance. It is then compatible with processing using the Zero-Mean Non-Linear (ZMNL) function to impart a non-Gaussian character. Its variance is subsequently corrected to obtain a signal that matches the target PSD.

The ZMNL function is the nonlinear transformation at the core of kurtosis control. It relies on the Hermite polynomial model [20]. This is defined as follows.

$$y = h[z + a(z^2 - 1) + b(z^3 - 3z)]$$

where y is the non-Gaussian excitation signal obtained from the Gaussian signal z with unit variance. The coefficients a , b , and h of the cubic function are directly related to the skewness and target kurtosis by the following formulas.

$$a = \frac{S}{4 + 2\sqrt{1 + 1.5(K - 3)}}$$

$$b = \frac{\sqrt{1 + 1.5(K - 3)} - 1}{18}$$

$$h = \frac{1}{\sqrt{1 + 2a^2 + 6b^2}}$$

The so-called "direct" DBM, dedicated to synthesis excitation, which does not rely on an intermediate distribution, is then applied to the non-Gaussian signal thus obtained. For this, the temporally generated blocks must be sufficiently long to satisfy the assumption of asymptotic convergence towards EVD for the MRS and Gaussian for the FDS. The duration must be chosen so as to get more than several hundred periods of the current dof. In addition, the signal's sampling frequency must be at least 30 times greater than the frequency of the current dof. What's more, the spectral content after filtering must be correctly represented (a loss of 5 % is accepted in our case). The Z-transform [17] and the Rainflow counting method [8] are used to calculate the MRS and FDS for each n-sample. Statistical inference is then performed based on the obtained distributions. Since the response signal is stochastic in nature, the response spectra calculated in this way are associated with a quantile characterized by a given exceedance risk.

The entire process of generating non-Gaussian excitation signals and determining their response spectra is summarized in figure 7.

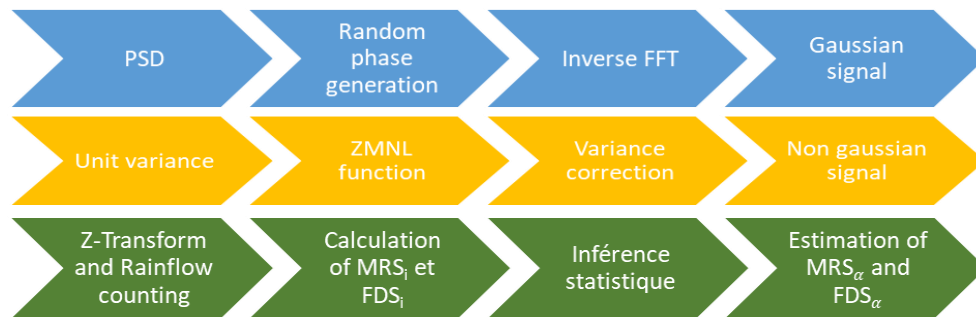


Figure 7 Process of generating non-Gaussian signals [19].

An example of generation is given below. The previous PSD (see §3.2) is used, and the time-domain signals obtained for different kurtosis values are presented in figure 8.

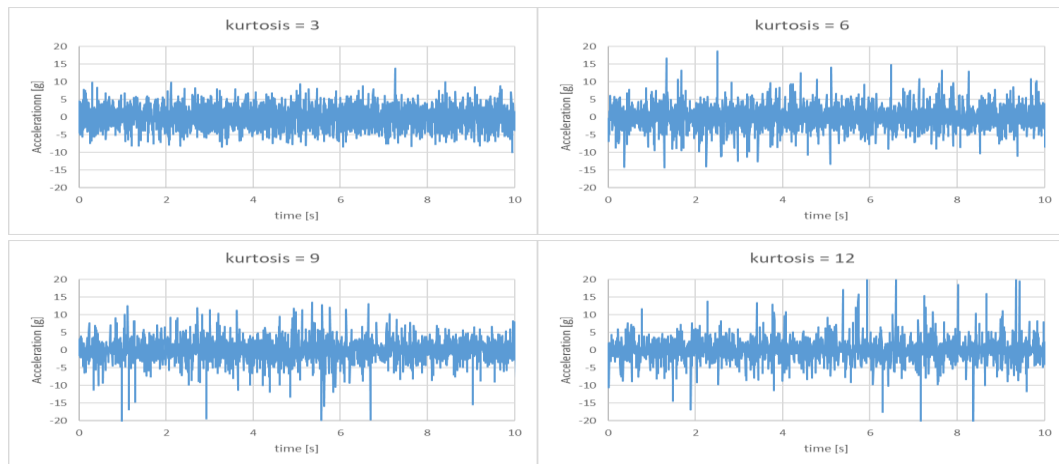


Figure 8 Time-domain signals generated from the PSD of random noise at $0,1 \text{ g}^2/\text{Hz}$ from 1 to 400 Hz for $k = [3 ; 6 ; 9 ; 12]$.

5 Results and discussion

5.1 Data classification by spectral similarity

To illustrate the relevance of the proposed classification, real vertical acceleration signals, measured on the roof of a wheeled vehicle in motion on a path, are used. These signals have a sampling frequency of 5 kHz. They are divided into temporal blocks of 0.8192 seconds in duration, corresponding to a buffer size of 4096 points, to be compatible with numerical processing via the fast Fourier transform. The duration, close to one second, allows for the assumption of stationarity within the block while ensuring that it contains enough data to be robust. The blocks are then classified into stationary clusters based on the similarity of their spectral content using the introduced method. For this, the SSI threshold value is set to 85 %. The following table lists the formed clusters and their size after the classification of the temporal blocks in the considered driving situation.

Table 1 Synthesis of the classification of data from the gravel road driving situation.

Cluster	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Size	508	253	55	195	59	16	8	60	73	144	6	27	53	8	238	5	5	9	4	2

The Amplitude AutoSpectra (ASA) of the 16 temporal blocks grouped within the sixth cluster are plotted in figure 9. The overlap of the spectra highlights the stationarity of the temporal blocks from a spectral content perspective. This stationarity and the coherence of the data are essential prerequisites for filtering analysis using single-dof resonators.

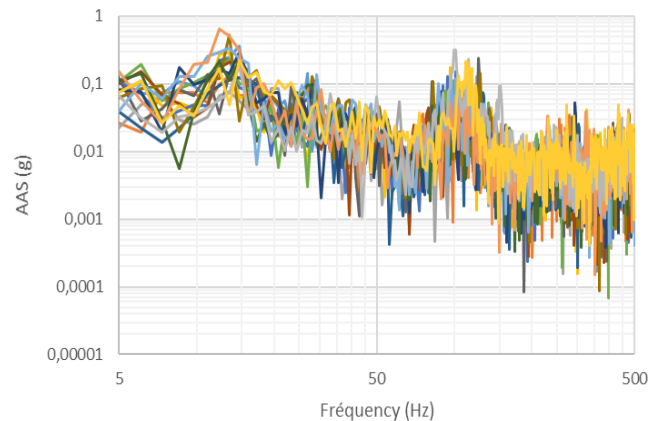


Figure 9 Amplitude AutoSpectra (ASA) of time blocks grouped within the sixth cluster.

5.2 Study of Gaussian random noise

The validity of the presented formalization method is first evaluated by studying first a Gaussian random noise. For this, the Gaussian random signal generated from the previous PSD (see §3.2) is used. The MRS at the exceedance risk α is then calculated analytically from the PSD for a t_p duration of 10 000 hours, deterministically from the generated time-domain signal. It is also calculated using the indirect DBM, through the Kappa distribution and the generalized Rayleigh distribution, with a temporal block size of 1 second and a t_p duration of 10 000 hours, corresponding to an extrapolation factor of 3.6×10^7 . The calculation parameters are as follows: $Q = 10$, $C = K = 1$. The MRS, with an exceedance risk $\alpha = 10\%$, over the frequency range [1; 500] Hz, with a resolution of 1/24 octave, are plotted in figure 10.

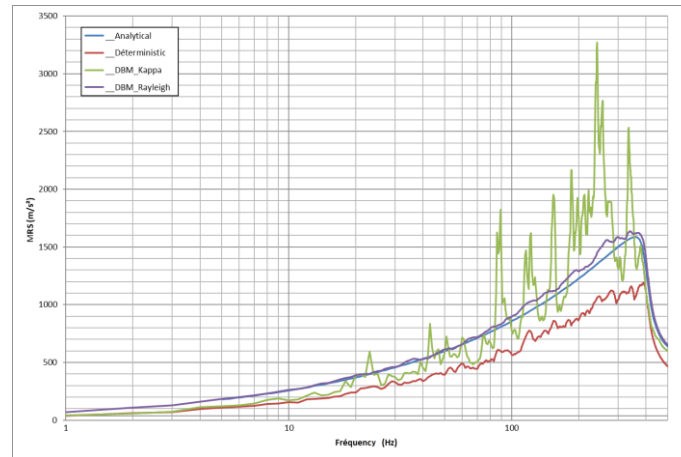


Figure 10 Analytic MRS via direct DBM (blue), determinist MRS (red), MRS obtained via the indirect DBM with the Kappa distribution (green) and the generalized Rayleigh distribution (purple) with an exceedance risk $\alpha=10\%$.

The analytically calculated MRS serves as an exact reference in this case, where the studied signal is Gaussian. The MRS calculated using the deterministic method is strictly lower than the analytically calculated MRS. The inability of the deterministic method to predict the stochastic nature of extreme event occurrences is well confirmed [1].

The MRS calculated via the indirect DBM and the Kappa distribution shows significant fluctuations between different degrees of freedom. A maximum error of 135 % compared to the deterministic MRS is observed. This phenomenon is attributed to the many possible distributions for the convergence of the statistical model. It is highlighted on the $(\tau_3 ; \tau_4)$ diagram in figure 11 by the dispersion of the n-samples on either side of the $k=0$ curve. Thus, for the different degrees of freedom, the statistical models converge towards distributions with different abilities to infer extreme events: Weibull ($k>0$), Gumbel ($k=0$), and Fréchet ($k<0$). In the case of Gaussian signals, the Kappa distribution is therefore not suitable due to strong fluctuations between different frequencies, thus generating an error incompatible with the prediction of response spectra.

The MRS calculated via the indirect DBM and the generalized Rayleigh distribution shows, as previously observed, a strong correlation with the analytically calculated MRS. The spectrum exhibits very few fluctuations, of low amplitude, between the degrees of freedom despite a high extrapolation factor. This stability is allowed by a single statistical distribution, derived from the field. The generalized Rayleigh distribution therefore shows good potential for performing high-quality inference and precise extrapolation.

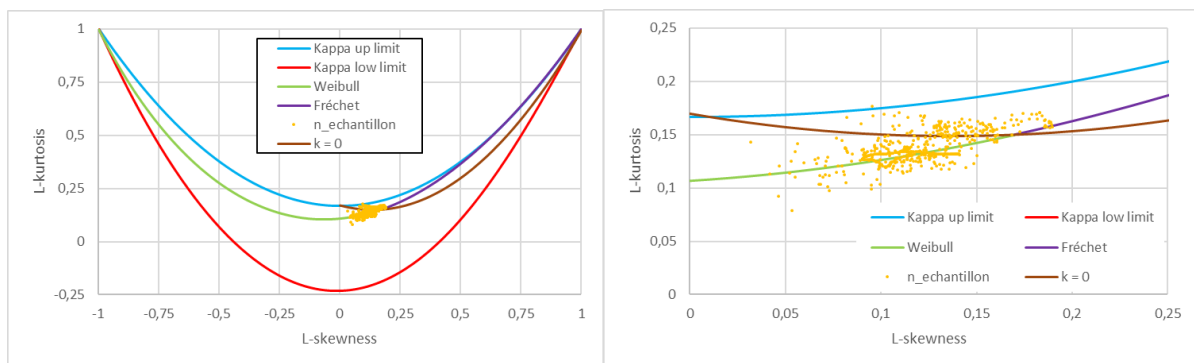


Figure 11 Diagram of the L-moment ratio $(\tau_3 ; \tau_4)$ for different statistical models and the n-samples of the Gaussian signal (yellow points), over the entire domain (left) and detailed view (right).

5.3 Study of non-Gaussian signals

The implemented methods are now evaluated for the study of non-Gaussian signals. To do this, signals are generated following the process described in section 4.3. The PSD of 0.1 g²/Hz from 1 to 400 Hz (see §3.2) and the kurtosis values $k = [3; 6; 9]$ are used. The MRS of the generated signals are determined via the direct DBM and plotted in figure 12 with an exceedance risk of 10 %. The calculation parameters for the MRS remain unchanged compared to section 5.2.

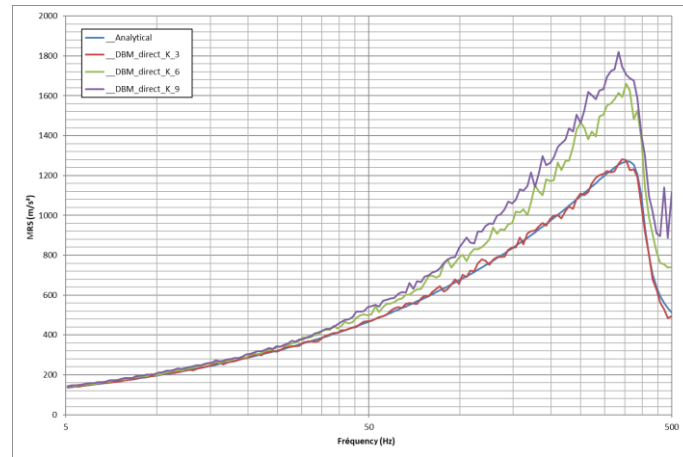


Figure 12 Analytical MRS (blue), MRS obtained via the direct DBM with a kurtosis $k=3$ (red), $k=6$ (green) and $k=9$ (purple) with an exceedance risk $\alpha=10\%$.

The MRS show oscillations that become more pronounced as the kurtosis increases. This phenomenon is attributed to the increase in the coefficient of variation, the ratio of the standard deviation to the mean, along with the kurtosis. Indeed, for the signal generated with $k=3$, the coefficient of variation is zero, while it is 30 % for the signal generated with $k=9$. A signal leak is also observed beyond the frequency range defined by the target PSD, starting from 400 Hz. Thus, the generated temporal blocks take on spectral values that can vary between the different degrees of freedom, resulting in noise in the MRS. These disturbances do not prove detrimental to the present study, as the stationarity of the generated signals is guaranteed by the first four statistical moments.

The MRS are also calculated from the generated time-domain signals, with an exceedance risk of 10 %, using the indirect DBM with the Kappa and generalized Rayleigh distributions. They are plotted in figure 13. The extrapolation is performed over a duration of 1 hour.

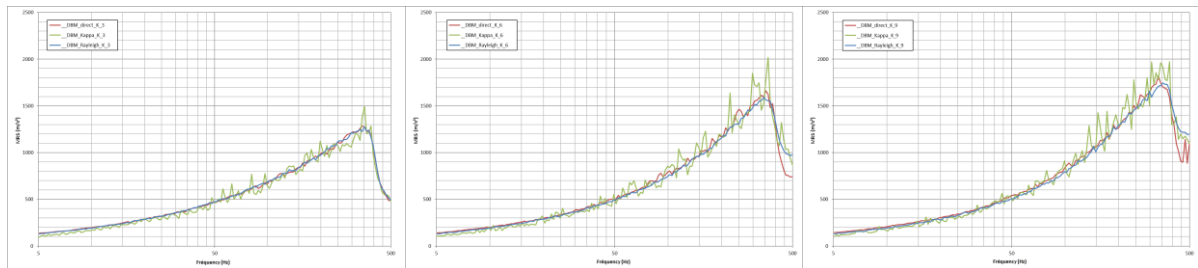


Figure 13 MRS obtained via the direct DBM (red), via the indirect DBM with the Kappa distribution (green) and the generalized Rayleigh distribution (blue), for signals generated with a kurtosis of 3 (left), 6 (center) et 9 (right).

The MRS obtained via the indirect DBM and the Kappa distribution show significant oscillations. A discrepancy of about 30 % is observed compared to the MRS determined via the direct DBM, despite a significantly reduced extrapolation factor compared to the Gaussian study case (3.6×10^3 vs 3.6×10^7).

Similarly, this phenomenon is due to the convergence towards different statistical distributions across the frequency range, including between successive degrees of freedom. Indeed, the $(\tau_3 ; \tau_4)$ diagram in figure 14 again illustrates that the n-samples from the non-Gaussian signals, even if they are mainly on the left side of the $k=0$ curve, are spread on either side of the $k=0$ curve. The proximity of the non-Gaussian n-samples to the domains covered by the extreme value laws is observed. This shows the convergence of the generated temporal blocks and highlights the relevance of applying the direct DBM.

The MRS determined via the indirect MBD and the generalized Rayleigh distribution again show a strong correlation with the MRS obtained by direct DBM. This stability is explained by the asymptotic convergence towards a single statistical distribution. It thus offers a simplified implementation of the DBM, allowing for comparison in inter-laboratory studies.

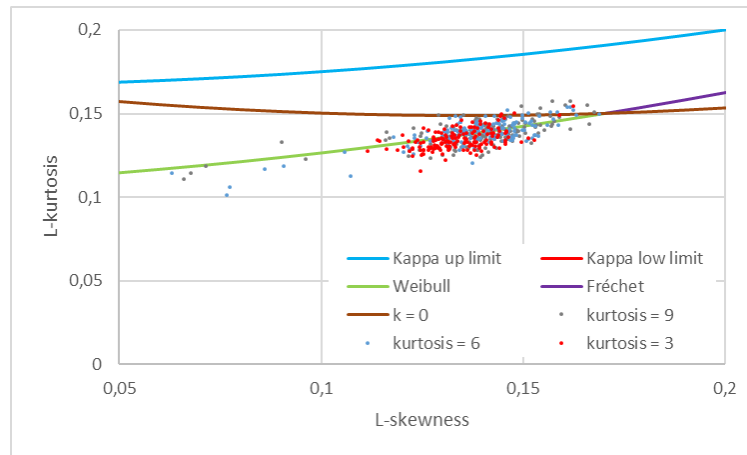


Figure 14 Diagram of the L-moment ratio $(\tau_3 ; \tau_4)$ for the different statistical models and the n-samples of the generated signals for $k=3$ (red points), $k=6$ (grey points), and $k=9$ (blue points).

6 Conclusion

This publication proposes major innovations in the methodology for considering mechanical environments, as defined by the AFNOR X50-144-3 standard. A new approach to data classification is implemented. The robustness of this method, based on a spectral similarity criterion, is an essential prerequisite for processing data via filtering through single-degree-of-freedom resonators. In order to model the stochastic nature of the system's response in terms of extreme stress, the Kappa and generalized Rayleigh distributions are introduced. Their ability to infer statistical distributions is evaluated by applying them to the processing of data of various types. On one hand, in the case of Gaussian signals, the response spectra estimated via the newly introduced statistical models are compared to those calculated analytically. On the other hand, non-Gaussian signals, whose effects in terms of response spectrum are known, are generated. These serve as a reference for evaluating the models in the formalization method for this type of vibratory phenomena. In the cases studied, the Kappa distribution proves to be unsuitable for inference, with consistency between the degrees of freedom. In contrast, the Rayleigh distribution demonstrates the ability to infer both Gaussian and non-Gaussian signals while limiting fluctuations across the frequency range. It also offers simplicity in implementing the formalization via the Disjoint Blocks Method with a single "a priori" distribution, thus enabling reproducibility and inter-laboratory study comparisons.

Finally, the inference method based on Rayleigh's law proves to be a simple and efficient solution for vibration analysis. However, it should be noted that specific adaptations may be necessary for

particular situations, such as shocks or excitations from rotating machines. In these cases, additional approaches, such as sine on random noise processing, may be relevant.

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